

KAZAN, Russian FederationWMO: **275950**Lat: **55.7400N**Lon: **49.2044E**Elev: **389**StdP: **14.49**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.7	-11.5	-23.7	1.5	-17.5	-17.2	2.2	-11.0	27.0	19.4	24.5	22.1	7.5	140	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	88.0	67.6	84.4	66.7	80.9	65.2	70.4	82.0	68.8	80.0	67.2	77.9	8.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.4	98.6	75.7	64.7	92.7	73.6	63.0	87.2	72.4	34.6	82.1	33.2	80.1	31.9	77.7	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	23.7	21.4	19.7	-21.7	91.8	6.0	3.6	-26.0	94.4	-29.5	96.5	-32.9	98.5	-37.2	101.2	
(5)				-22.0	73.5	5.9	2.4	-26.3	75.3	-29.7	76.7	-33.1	78.1	-37.4	79.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.5	12.5	12.8	24.2	41.8	56.2	64.0	68.3	64.8	54.1	40.9	27.2	16.9	
	DBStd	22.35	12.73	12.53	8.54	9.68	9.14	7.74	6.65	7.63	7.90	7.83	10.18	12.43	
	HDD50	5383	1162	1040	798	284	43	4	0	1	45	296	684	1026	
	HDD65	9304	1627	1460	1263	697	297	110	42	98	339	746	1134	1491	
	CDD50	1906	0	0	0	37	236	422	568	461	167	15	0	0	
	CDD65	352	0	0	0	0	26	79	145	92	10	0	0	0	
	CDH74	3155	0	0	0	4	270	704	1276	832	69	0	0	0	
	CDH80	969	0	0	0	0	53	209	406	292	9	0	0	0	
Wind	WSAvg	9.2	10.0	9.7	9.7	9.3	9.5	8.5	7.6	8.3	8.6	9.6	9.8	9.8	
Precipitation	PrecAvg	21.2	1.5	1.2	1.2	1.3	1.4	2.6	2.6	2.4	2.0	1.9	1.7	1.6	
	PrecMax	31.4	3.4	3.2	3.4	3.7	3.8	8.8	7.7	5.6	4.5	5.4	4.3	3.6	
	PrecMin	14.6	0.2	0.0	0.1	0.2	0.1	0.3	0.1	0.0	0.2	0.0	0.2	0.5	
	PrecStd	4.0	0.7	0.7	0.8	0.7	0.9	1.7	1.3	1.4	1.2	0.9	0.8	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.5	36.0	44.1	73.7	85.7	90.1	92.8	93.5	80.7	65.3	49.8	39.1	
		MCWB	34.1	33.7	38.0	55.9	63.6	68.0	66.9	68.1	62.3	53.6	48.0	38.0	
	2%	DB	33.8	34.1	39.5	68.1	80.7	85.9	87.9	87.5	75.3	59.2	45.3	35.8	
		MCWB	32.8	32.9	35.3	52.1	61.8	67.2	68.1	67.1	59.9	51.1	43.3	34.5	
	5%	DB	32.1	32.7	37.2	62.8	76.8	81.5	84.4	82.4	71.0	55.4	41.8	34.0	
		MCWB	31.3	31.9	34.0	50.7	59.8	65.7	67.5	65.8	58.6	49.4	40.4	32.9	
	10%	DB	30.0	30.4	35.3	57.2	71.9	77.4	80.9	78.3	66.3	52.0	38.8	32.3	
		MCWB	29.3	29.4	33.0	47.8	57.3	64.0	66.3	64.2	57.0	47.7	37.6	31.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.1	34.2	38.7	57.4	66.0	71.9	73.1	71.1	64.8	56.0	47.8	37.5
			MCDB	34.6	34.9	43.0	70.2	80.1	84.0	83.6	83.9	75.2	61.6	49.3	38.2
2%		WB	32.6	33.0	35.8	54.3	63.5	69.3	71.0	69.1	62.0	52.4	43.6	34.6	
		MCDB	33.2	33.7	38.5	65.6	77.4	81.1	82.2	81.8	71.4	57.6	44.8	35.3	
5%		WB	31.2	31.9	34.4	51.2	61.2	67.4	69.4	67.3	59.7	50.2	40.3	32.9	
		MCDB	31.9	32.7	36.2	61.5	73.0	78.4	80.0	79.4	68.1	53.9	41.4	33.7	
10%		WB	28.8	29.3	33.1	48.1	59.0	65.5	67.7	65.6	57.6	48.3	37.3	31.4	
		MCDB	29.6	30.3	34.7	57.2	69.4	75.4	77.6	76.0	65.2	51.8	38.2	32.2	
Mean Daily Temperature Range	5% DB	MDBR	11.0	12.4	13.0	15.6	18.6	17.8	17.8	17.6	15.0	10.7	8.2	10.0	
		MCDBR	8.2	9.6	12.8	23.0	23.9	22.7	22.1	23.2	21.3	16.4	10.1	8.4	
		MCWBR	8.2	9.1	9.5	12.1	10.6	10.0	9.0	9.1	10.0	10.2	9.1	7.9	
	5% WB	MCDBR	8.5	9.2	11.3	21.9	21.9	20.2	19.4	20.1	18.9	14.3	9.1	8.0	
		MCWBR	8.5	8.9	8.9	12.4	10.8	10.0	9.1	8.8	10.2	9.9	8.7	8.0	
Clear-Sky Solar Irradiance	taub	0.290	0.318	0.362	0.408	0.388	0.380	0.401	0.428	0.375	0.349	0.312	0.286		
	taud	2.061	2.049	2.010	2.120	2.321	2.387	2.334	2.234	2.362	2.327	2.188	2.043		
	Ebn at Noon	185	228	247	255	271	274	265	247	246	221	179	153		
	Edh at Noon	21	31	41	43	37	35	36	38	29	23	18	16		
All-Sky Solar Radiation	RadAvg	216	514	965	1351	1749	1879	1837	1444	883	414	168	120		
	RadStd	31	73	110	114	127	171	163	112	109	68	31	30		

Historical Trends

		DBAvg	Heating		Cooling
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